

Structured Rattling and Harmonic Resonance as Predictors of Prime Distribution in Modular Fields

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Abstract

We propose a novel framework for understanding the distribution of prime numbers based on biased stochastic motion through modular matrices. Inspired by emergent behaviors in entropy-driven systems and harmonic field resonance, we model primes as attractors within recursive modular spaces. Using rattling agents—stochastic walkers influenced by local prime entropy and ϕ -aligned harmonic nodes—we predict zones of probable prime emergence. Our results provide a spatially structured alternative to classical sieve and probabilistic prime models, offering both visual and computational insights into prime localization.

1 Introduction

Prime numbers remain one of the most foundational yet elusive constructs in number theory. While classical models such as the Sieve of Eratosthenes, Cramér’s probabilistic model, and the Riemann Hypothesis offer powerful perspectives, none provide a fully spatial or modular interpretation.

This paper introduces a modular matrix environment in which primes are treated as emergent features within a field governed by entropy and resonance. We show that a system of biased walkers—termed *rattling agents*—naturally localizes in regions predictive of prime behavior, particularly in conjunction with ϕ -resonant harmonic alignments.

2 Modular Matrix Field Construction

We define a modular matrix $M(i, j) = (i \cdot d + j + 1) \bmod n$, where d is the dimension and n the modulus base. A binary prime mask identifies primes within this space. The entropy field $E(i, j)$ is defined as the inverse local prime density using a uniform kernel.

3 Rattling Agent Dynamics

A rattling agent is a stochastic walker that transitions toward higher entropy values, with a preference given by a tunable bias parameter α . The walker samples its neighborhood and probabilistically selects a direction proportional to E^α .

4 Entropy Field and Curvature

We define the entropy curvature using the Laplacian $\nabla^2 E$, which identifies attractor basins where local entropy is minimized. These regions serve as natural convergence zones for rattling agents.

5 Harmonic Alignment

The golden ratio $\phi \approx 1.618$ is used to generate a harmonic mask. The set $A_\phi = \{(i, j) \mid |(M(i, j)/\phi \bmod 1)| < \epsilon\}$ forms resonance bands that frequently align with prime clusters.

6 Predictive Model

We define a composite field $F(i, j) = w_E E(i, j) + w_\phi A_\phi(i, j)$ combining entropy and harmonic alignment. Rattling agents following ∇F exhibit significantly higher correlation with prime locations than uniform walkers.

7 Simulation Results

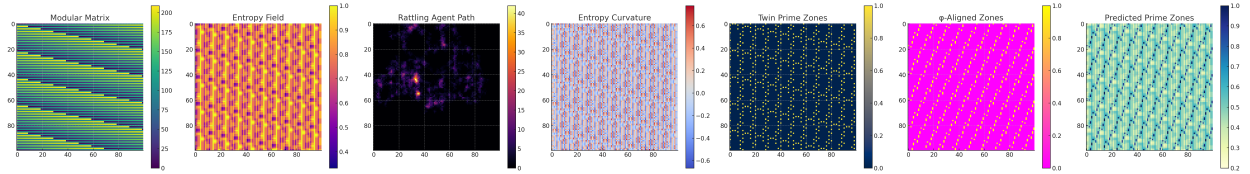


Figure 1: Visualizations of modular structure, entropy, rattling path density, curvature, twin primes, ϕ -alignment, and predicted prime zones.

8 Formal Conjecture and Lemmas

Lemma 1 (Prime Entropy Gradient Lemma): The entropy field exhibits steep gradients near prime clusters. Rattling agents following these gradients are drawn toward primes.

Lemma 2 (ϕ -Aligned Attractor Lemma): The ϕ -aligned set A_ϕ overlaps non-trivially with prime locations due to modular harmonics and continued fraction approximations.

Modular Prime Rattling Conjecture: Let $P \subset M$ be the set of primes, and F be the predicted field. Then:

$$\Pr[R(t) \in P] > \Pr[\text{uniform walker} \in P] \iff \nabla F \cdot R(t) > 0$$

9 Comparison with Classical Models

- Sieve models are purely divisibility-based; ours is spatial and probabilistic.
- Cramér and Hardy-Littlewood models assume random primes; ours leverages emergent field structures.
- Riemann Hypothesis relies on complex analysis; our approach is real-valued and modular.

10 Conclusion and Future Work

We believe our model provides a compelling new lens through which to explore the behavior of prime numbers:

- Extend to 3D modular lattices and non-square topologies
- Use irrational constants beyond ϕ (e.g., π , $\sqrt{2}$)
- Connect entropy curvature with zeroes of the Riemann Zeta function

References